Juniper Core Node BLR-CORE-01 Failure SOP (ALARM_6778)

Node: BLR-CORE-01 | **Vendor:** Juniper | **Severity:** Warning | **Alarm Code:** ALARM_6778

Description

This Standard Operating Procedure (SOP) outlines the steps for troubleshooting and resolving hardware/environmental faults on the Juniper Core node BLR-CORE-01, which is experiencing ASIC memory parity errors, line card crashes, and power supply redundancy loss. The Juniper MX Series routers are designed with a modular architecture, where each line card or switching fabric module operates independently, yet in conjunction with other components to provide high-bandwidth, low-latency forwarding. However, due to the complex interplay of hardware components, software drivers, and environmental factors, failures can occur. This SOP focuses on identifying and resolving the root cause of ALARM_6778 on the specified node, which falls under the hardware/environmental fault domain.

Pre-requisite Checks

- pre-check: Verify the site location and ensure remote hands are dispatched to the Unknown Site for physical access and assistance.

- pre-check: Confirm the status of redundant links and ensure they are operational.

- pre-check: Check the power supply redundancy state on the node and ensure it's functioning as expected.

Potential Root Causes

Cause	Probability
ASIC memory parity error due to hardware degradation or manufacturing defect.	High
Line card crash or misconfiguration leading to packet loss and forwarding issues.	Medium
Power supply redundancy loss due to hardware failure, misconfiguration, or environmental factors.	Low

Diagnostic Commands & Expected Outputs

```
Command: show chassis hardware, Expected Output: Chassis hardware, Type: MX960, Serial #: XXXXXXXX, Location: BLR-CORE-01, Description: This command displays the node's chassis hardware details, including the serial number and location, helping to identify potential hardware issues.

Command: show chassis environment, Expected Output: Temperature: 25 degrees C, Fan speed: 30%, Power supply: operational, Description: This command shows the node's environmental factors, such as temperature and fan speed, aiding in identifying potential overheating or cooling issues.
```

```
Command: show chassis power, Expected Output: Power supply 1:
operational, Power supply 2: operational, Redundancy: active/standby,
Description: This command displays the power supply status, including
redundancy, to ensure it's functioning as expected.
```

Resolution Steps

Step 1: Step: Replace faulty line card or switching fabric module., Command: request chassis replace linecard , Description: Replace the faulty line card or switching fabric module with a new one, ensuring the node's forwarding capabilities are restored.

Step 2: Step: Reconfigure power supply redundancy., Command: set chassis power redundancy , Description: Reconfigure the power supply redundancy mode to ensure the node operates in the desired redundancy state.

Step 3: Step: Update ASIC firmware., Command: request system software add , Description: Update the ASIC firmware to the latest version, addressing potential hardware or software issues causing the alarm.

Post-Check Verification

```
Command: show chassis hardware, Expected Output: Chassis hardware,
Type: MX960, Serial #: XXXXXXXX, Location: BLR-CORE-01, Description:
Verify the node's chassis hardware details to ensure the replacement
or upgrade was successful.

Command: show chassis environment, Expected Output: Temperature: 25
degrees C, Fan speed: 30%, Power supply: operational, Description:
Verify the node's environmental factors to ensure they are within the
normal operating range.

Command: show chassis power, Expected Output: Power supply 1:
operational, Power supply 2: operational, Redundancy: active/standby,
Description: Verify the power supply status to ensure it's functioning
as expected.
```

Roll-back Procedures

- Step: Restore original configuration., Command: load configuration , Description: Restore the original configuration to revert any changes made during the resolution process.

- Step: Replace original hardware components., Command: request chassis replace linecard , Description: Replace the original hardware components to restore the node to its previous state.

- Step: Revert ASIC firmware to original version., Command: request system software delete , Description: Revert the ASIC firmware to its original version, undoing any changes made during the resolution process.